Part A [15 marks]

Question 1 [15 marks]

(a) [5 marks] Questions (i)-(v) are each worth one mark and have a single correct answer. If you believe more than one answer is correct, you should still only select one answer.

(i) What term describes a difference in attributes between the target population and sample?

A Study error

B Sample error

C Measurement error

D None of the above

(ii) We should check the fit of a statistical model during which step of PPDAC?

A Problem

B Plan

C Data

D Analysis

(iii) Which of these statements is TRUE?

A A point estimate is a constant.

B A point estimate is a random variable.

C We never know the true value of a point estimate.

D More than one of the above statements is true.

(iv) A 10% likelihood interval for an unknown parameter θ is $[1, 2]$. Which of the following statements is TRUE?

A We are 10% confident $\theta \in [1, 2]$

B $P(1 \leq \theta \leq 2) = 0.1$

C Both A and B are true.

D Neither A nor B is true.

(v) When testing $H_0 : \sigma^2 = \sigma_0^2$ for a $G(\mu, \sigma)$ model with μ, σ unknown and a sample of size n, which probability distribution should we use in the calculation of the p-value?

A $G(0, 1)$

B t_{n-1}

C χ_1^2

D χ_{n-1}^2

(b) [5 marks] Attendees at a party are offered either an alcoholic or non-alcoholic drink. Let $Y \sim \text{Binomial}(n, \theta)$ denote the number of people who choose an alcoholic drink in a random sample of n party attendees.

In a random sample of 40 party attendees, 30 chose an alcoholic drink.

You are given the following R commands and output:

```
> qt(0.95, 39)
[1] 1.684875
> qt(0.95, 40)
[1] 1.683851
> qnorm(0.95)
[1] 1.644854
```

```
> qt(0.975, 39)
[1] 2.022691
> qt(0.975, 40)
[1] 2.021075
> qnorm(0.975)
[1] 1.959964
```

```
> qt(0.995, 39)
[1] 2.707913
> qt(0.995, 40)
[1] 2.704459
> qnorm(0.995)
[1] 2.575829
```

(i) [1 mark] Write down a numeric expression for the maximum likelihood estimate of θ.

$\frac{30}{40} = 0.75$ [1 mark]

(ii) [2 marks] Construct an approximate **99%** confidence interval for θ.

The CI is given by $\hat{\theta} \pm a\sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}}$ where $\hat{\theta} = \frac{30}{40} = 0.75, n = 40, a = 2.575829$. This gives [0.573, 0.926]

[1 mark for correct equation $\hat{\theta} \pm a\sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}}$. 1 mark for correct quantile.]

(iii) [2 marks] A new study is proposed which will sample $n = 50$ party attendees. What is the maximum possible width of a 99% confidence interval that could result from this new study? Note: This is separate to the data mentioned earlier in the question.

Maximum width $= 2a\sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}}$ where $n = 50, a = 2.575829, \hat{\theta} = 0.5$. This gives 0.364.

1 mark for correct equation, 1 mark for noting it is widest when $\hat{\theta} = 0.5$

[Note: While we do not know $\hat{\theta}$ for this new study, the interval will be widest when $\hat{\theta} = 0.5$.]

(c) [5 marks] A grocery store sells butter tarts which are each advertised to weigh 50 grams. Amar is hosting a STAT 231 Midterm 2 study party and buys 49 tarts. At the party, Amar's friends decide it would be super fun to weigh all of the tarts to test the hypothesis that the advertised weight is accurate. (Amar thinks this is a bit strange and just wanted to have a nice snack, but decides to go along with it anyway.)

The friends find that the sample mean weight is 52 grams with a sample standard deviation of 7.

Let $Y \sim G(\mu, \sigma)$ denote the weight of a randomly chosen butter tart. Consider the discrepancy measure $D = \frac{|\overline{Y} - E[Y;H_0]|}{S/\sqrt{n}}$ where $\overline{Y}$ is the estimator of the sample mean and S is the estimator of the sample standard deviation.

(i) [1 mark] Write down the null hypothesis H_0 and give the numeric value of $E[Y; H_0]$ (the expected value of Y if the null hypothesis is true).

$H_0 : \mu = 50 \qquad E[Y; H_0] = 50$

0.5 marks each.

(ii) [2 marks] Give a probabilistic expression for the p-value resulting from the hypothesis test using the discrepancy measure D and the null hypothesis you gave in (i). Your answer should clearly include a numeric value for d.

$d = \frac{|52-50|}{7/\sqrt{49}} = 2.$

$p - \text{value} = P(D \geq d; H_0) = P(D \geq 2) = P(T < -2) + P(T > 2)$ where $T \sim t_{48}$

0.5 marks for correct value of d. 0.5 marks for correct distribution t_{48}. 1 mark for correct probabilistic expression.

(iii) [1 mark] You are told that the p-value resulting from the hypothesis test in part (ii) is less than 0.05. Would a 90% confidence interval for μ based on these data contain the value 50? Answer with either 'Yes', 'No', or 'Not enough information/NEI'.

No.

Note: A 90% confidence interval is narrower than a 95% interval, and we know that a 95% interval would not contain 50, so neither would a 90% interval.

(iv) [1 mark] Amar realizes each butter tart was weighed in its container, which added 1 gram to the weight of each butter tart. If they repeated the hypothesis test in part (ii) ignoring the weight of the containers, would the p-value increase, decrease, or stay the same?

Increase.

Note: The new sample mean would be 51 grams, which is closer to the hypothesized value of 50 grams. Nothing else in the analysis would change, so the evidence against the null would weaken, and the p-value would increase.

Part B [15 marks]

Part B begins on the following page and are based on the assignments. All variate names refer to those in the Assignment Dataset Information file. All R commands assume the assignment dataset has been stored in R in an object named `mydata`. You may further assume that the variate `subject.age.log` has been created (as a variate within `mydata`) as described in Analysis 2 of Assignment 1, and that the MASS library has been loaded.

Question 2 [15 marks]

(a) [8 marks] All questions in this part concern Assignment 2. Any references to specific analyses, or models, are in relation to those detailed in Assignment 2.

Questions (i)-(v) are each worth one mark and have a single correct answer. If you believe more than one answer is correct, you should still only select one answer.

(i) Which R command would construct a relative frequency histogram of `subject.age`?

A `hist(mydata$subject.age)`

B `truehist(mydata$subject.age)`

C Both A and B would accomplish this.

D Neither A nor B would accomplish this.

(ii) What is the range of $R(\theta)$ as defined in Analysis 1h?

A [0, 1] [Note: the range of $R(\theta)$ corresponds to the y-axis.]

B [28, 41]

C [0, 34.56]

D It depends on the value of θ

(iii) Let $A \sim \text{Exponential}(\theta)$ as defined in Analysis 1. What is the maximum likelihood estimate of the standard deviation of A?

A $\sqrt{34.56}$

B 34.56 [Note: This is the sample mean, via the invariance property.]

C 34.56^2

D $\hat{\sigma}$

(iv) Which of the following CANNOT be estimated from the Q-Q plot generated in Analysis 2f?

A Sample median value of `vehicle.year` for your chosen city.

B Interquartile range of `vehicle.year` for your chosen city.

C Maximum value of `vehicle.year` for your chosen city.

D All of A-C can be estimated from the Q-Q plot.

(v) Which of the following R commands would create a side-by-side boxplot of `vehicle.year` within each category of `vehicle.make`?

A `boxplot(mydata$vehicle.year, mydata$vehicle.make)`

B `boxplot(mydata$vehicle.make, mydata$vehicle.year)`

C `boxplot(mydata$vehicle.year ~ mydata$vehicle.make)`

D `boxplot(mydata$vehicle.make ~ mydata$vehicle.year)`

Questions (vi)-(viii) are each worth one mark. Consider each of the following statements based on Assignment 2. For each statement, discuss whether you agree or disagree, providing a brief justification for your answer in 1-2 sentences.

(vi) “We do not have concerns about sample error in this analysis.”

Agree. The sample is a random sample of the study population, and so regardless of the atrribute under study, we do not have concernsa about sample error.

(vii) “Based on the relative frequency histogram in Analysis 1c, an Exponential model for this variate is not appropriate for `subject.age`.”

Agree. They are very different when we would expect them to be the same if the data were generated from an Exponential distribution.

(viii) “The expected frequencies calculated in Analysis 2a should sum to 950.”

Disagree. We are calculating the expected frequencies within a particular city, so they should sum to the number of stops in that city.

(b) [7 marks] All questions in this part concern Assignment 3. Any references to specific analyses, or models, are in relation to those detailed in Assignment 3.

Questions (i)-(v) are each worth one mark and have a single correct answer. If you believe more than one answer is correct, you should still only select one answer.

(i) Suppose we wish to use `uniroot()` to find the bounds of a 10% likelihood interval. Which of these statements is FALSE?

A `uniroot()` will find approximate values for the interval.

B We must run `uniroot()` at least twice.

C `uniroot()` will return the lower bound first.

D We must specify lower and upper bounds for `uniroot()` to search between.

(ii) Which of the following can we NOT do using only the relative likelihood function plot produced in Analysis 1c?

A Estimate a 10% likelihood interval for ω.

B Estimate the sample size.

C Estimate the maximum likelihood estimate of ω.

D All of A-C can be done using this plot.

(iii) In Analysis 2 a $G(\mu, \sigma)$ model is proposed for log-transformed subject age. What is the maximum likelihood estimate of μ?

A 0.36

B 3.47

C 13.3

D 34.6

(iv) If we modify our definition of 'weekend' to include stops that happened after 9pm on Fridays, what would happen to our maximum likelihood estimate of ω?

A It would decrease.

B It would stay the same.

C It would increase.

D This cannot be determined from the dataset.

(v) Which R command is used when calculating the 90% confidence interval for σ in Analysis 2e?

A `qnorm(0.9)`

B `qnorm(0.95)`

C `qchisq(0.9, 633)`

D `qchisq(0.95, 633)`

Questions (vi)-(vii) are each worth one mark. Consider each of the following statements based on Assignment 3. For each statement, discuss whether you agree or disagree, providing a brief justification for your answer in 1-2 sentences.

(vi) "If ω denotes the probability a randomly chosen stop happens at the weekend, then 0.9 is a more plausible value for ω than 0.1."

Disagree. The m.l.e. of ω is 0.210, which is closer to 0.1 than 0.9.

(vii) "We should be concerned about the validity of the Central Limit Theorem approximation when we estimate θ in the Exponential(θ) model in Analysis 2b."

Disagree. We have a large sample size (634) and so it is reasonable to assume the CLT should be a satisfactory approximation.

Part C [30 marks]

Question 3 [10 marks]

The Ontario provincial government conducted a study of the number of patients arriving at Ontario hospital emergency departments during the year 2022. Of specific interest was the distribution of mean number of patients per day for each hospital in the province that year. The researchers conducting the study were given a list of hospitals in the Greater Toronto Area. 30 hospitals were selected at random from this list. Each hospital tracked the number of patients who arrived each day in 2022, allowing them to calculate the mean number of patients per day admitted at that hospital during the year 2022. This mean (one for each of the 30 hospitals) was reported to the researchers. Each of the 30 hospitals reported their results (that is, there were no missing data).

(a) [3 marks] Carefully describe the target population and study population.

Target population:

Ontario hospitals/emergency rooms in 2022.

0.5 marks for Ontario (must be Ontario, not Toronto/GTA), 0.5 marks for hospitals/emergency rooms (must be hospitals/emergency rooms, not patients), 0.5 marks for 2022.

Study population:

The hospitals/emergency rooms in the GTA on the researchers' list in 2022.

0.5 marks for 'in the GTA', 0.5 marks for hospitals/emergency rooms, 0.5 marks for 2022.

(b) [2 marks] Carefully describe the sampling protocol for this study.

30 hospitals were chosen at random from the study population.

0.5 marks for '30', 1 mark for 'chosen at random' (or equivalent), 0.5 marks for 'from the study population' (or equivalent).

(c) [2 marks] Carefully describe a possible attribute of interest of the study.

An attribute of interest might be the average number of patients arriving per day across all Ontario hospitals in 2022.

1 mark for some kind of function of the variates, 1 mark for setting it in the context of a population/group of units. Note that there are many possible alternate answers here!

(d) [2 marks] Would study error be a concern in this study? Briefly explain why or why not.

If GTA hospitals received more patients than other hospitals in Ontario, this would mean the attribute of interest was higher in the study population than the target population, which would be study error.

1 mark for an answer that discusses a difference between the target and study populations, 1 mark for relating this specifically to an attribute of interest. Note: An answer that concludes that there is no concern about study error can also receive credit, as long as a reasonable argument is made that addresses the points covered by these two marks.

(e) [1 mark] Is this a(n) (1) estimation, (2) hypothesis testing, or (3) prediction problem? (Choose (1), (2), or (3).) No justification is required.

Estimation.

This is an estimation problem because we are interested in estimating the distribution of number of patients.

Question 4 [10 marks]

Note: This question spans 3 pages.

Alex and Blake are two students working on co-op at *piZZZa*, a pizza company specializing in late night deliveries. Every pizza the company delivers has a number of toppings, as requested by the customer. Blake has been asked to analyze the number of toppings customers order on their pizzas.

Blake created an object in R called `toppings`, which contains the number of toppings on each of a sample of $n = 50$ pizzas. Blake has emailed Alex some questions, and includes the following R output in their email:

```
> n <- length(toppings)
> thetahat <- mean(toppings)
> n*ppois(0, thetahat)
[1] 6.766764
> n*ppois(1, thetahat)
[1] 20.30029
> n*ppois(2, thetahat)
[1] 33.83382
> thetahat - qnorm(0.95)*thetahat/sqrt(n)
[1] 1.534765
```

You may assume the number of toppings on each pizza represent realizations of independent and identically distributed random variables $Y_1, Y_2, \ldots, Y_{50}$ with $Y_i \sim \text{Poisson}(\theta), i = 1, 2, \ldots, 50$. Let $\hat{\theta}$ denote the maximum likelihood estimate of θ based on the observed sample.

(a) [2 marks] Note: <u>numeric values</u> (not numeric expressions) to at least 1 decimal place required. Assuming that number of toppings follow a Poisson($\hat{\theta}$) model, give the expected number of pizzas (out of 50) with:

(i) 0 toppings: 6.766764

(ii) 2 toppings: 13.53353

`ppois(2, thetahat)` $= P(Y \leq 2; \hat{\theta})$, so to find $P(Y = 2; \hat{\theta})$ we find the difference between `ppois(2, thetahat)` and `ppois[1, thetahat)`.

At the end of Blake's code they tried to calculate the lower bound of an approximate 99% confidence interval for θ based on a Central Limit Theorem approximation. Here is the relevant line of code:

```
> thetahat - qnorm(0.95)*thetahat/sqrt(n)
```

(b) [2 marks] There are two mistakes in Blake's code. Identify each mistake and explain what would be correct instead. Note: Your answer can describe the mistakes (and corrections) in words and/or algebra or, if you choose, you may write out the corrected line of code, underlining where it differs to Blake's line of code.

Corrected code:

```
thetahat - qnorm(0.995)*sqrt(thetahat)/sqrt(n)
```

There are two mistakes: `qnorm(0.995)` (not 0.95) should be used for the quantile, and the square root of `thetahat` should be used.

The approximate 99% confidence interval for θ is [1.485, 2.515]. You are reminded that the maximum likelihood estimate of θ is denoted by $\hat{\theta}$.

(c) [2 marks] For each of the following statements indicate whether they are True, False, or if more information is required (NEI) to determine if they are True or False. You must ensure your answer is clearly marked.

(i) $\theta \in [1.485, 2.515]$: True False <u>NEI</u>

θ is unknown, so we do not know whether it is in the interval.

(ii) $P(1.485 \leq \theta \leq 2.515) = 0.99$: True <u>False</u> NEI

θ is unknown, but constant, so the probability it is in the interval is either 0 or 1.

(iii) $\hat{\theta} \in [1.485, 2.515]$: <u>True</u> False NEI

$\hat{\theta}$ must be inside the interval given how the interval is calculated.

(iv) $P(1.485 \leq \hat{\theta} \leq 2.515) = 0.99$: True <u>False</u> NEI

Because $\hat{\theta}$ is inside the interval, this probability is 1.

After Alex corrected Blake's code, Blake sent the following reply:

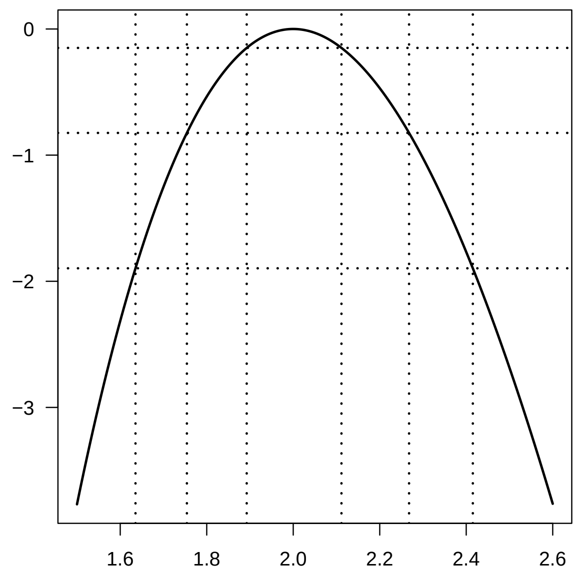
```
Thanks Alex, you're the best!! Also I told our manager that the Central Limit
Theorem approximation is reasonble for our data, because our sample size is bigger
than 40 - do you agree?
```

(d) [2 marks] Do you agree with Blake's statement that the Central Limit Theorem is reasonable because the sample size is larger than 40? Justify your answer.

While we may usually expect this to be a large enough sample size for the CLT to be reasonable, there is no reason to think that this is because the sample size is larger than 40. There is no specific sample size at which the CLT becomes reasonable.

Reminder: As we have discussed several times in class, the Central Limit Theorem does not become 'valid' at a certain sample size. Rather, it improves as the sample size increases. It also depends on other factors than the sample size, such as the shape of the distribution we are working with.

The following is a plot of the log relative likelihood function based on the Poisson(θ) model and the observed data. Horizontal lines have been added at -0.15, -0.824, and -1.897, along with intersecting vertical lines, for your guidance.



(e) [2 marks] Give a 15% likelihood interval for θ based on Blake's sample. Briefly explain how you found this interval. Your numeric values must be within 0.1 of the correct values.

[1.635297, 2.415254] This is found by looking at the horizontal line at $\log(0.15) = -1.89712$.

A 15% likelihood interval is the set of values θ such that $R(\theta) \geq 0.15$, therefore for a log relative likelihood function we find the values that satisfy $r(\theta) \geq \log(0.15)$, where log is to base e.

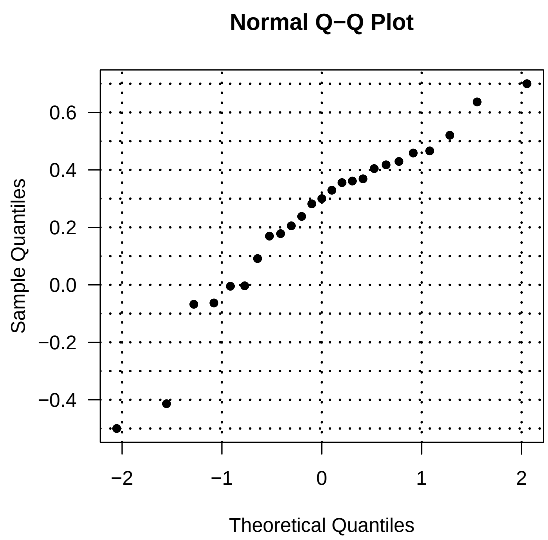
Question 5 [10 marks]

A study was designed to investigate the correlation between grades in math and physical education courses among high school students in Canada.

A study population of schools in British Columbia was formed. 25 schools were selected at random from this study population to participate in the study.

Each school was asked to take a random sample of 24 students, and for their sample compute the sample correlation between the students' grades in math and physical education. Each of the 25 schools reported the resulting sample correlation to the study.

These sample correlations are stored in R in an object named `r`. The following Gaussian Q-Q plot summarizes the data, where horizontal and vertical dotted lines have been added to aid interpretation.



(a) [2 marks] Give the following values (your answers must be within 0.1 of the correct value):

(i) The range of `r`: 1.2 or [-0.5, 0.7]

(ii) The sample median of `r`: 0.3

Reminder: The y-axis corresponds to our observed data, allowing us to read off the range, while the sample median is paired with 0 on the x-axis, as 0 is the median of a $G(0, 1)$ distribution.

Let Y = the correlation between math and physical education grades in a randomly chosen school from the study population. It is assumed that $Y \sim G(\mu, \sigma)$ with μ and σ both unknown.

The investigators wish to test the null hypothesis $H_0 : \mu = 0.1$. You are given the following R commands and output. Note that in the first line of the output below `t.test(r, mu = 0.1)` three numbers have been replaced by `(A)`, `(B)`, and `(C)`.

```
> mean(r)
[1] 0.2344359
> sd(r)
[1] 0.289478
> t.test(r, mu = 0.1)
t = (A), df = (B), p-value = (C)
alternative hypothesis: true mean is not equal to 0.1
```

(b) [2 marks] What value has been replaced by `(A)`? Leave your answer as a numeric expression or numeric value.

$\frac{|\bar{y}-\mu_0|}{s/\sqrt{n}} = \frac{0.2344359-0.1}{0.289478/\sqrt{25}} = 2.322$

Note: The sample size is 25 (not 24) because the dataset is of the correlation submitted by each of the 25 schools. While each school sampled 24 students, the dataset we are analyzing is of the school-level data, not the student-level data.

(c) [2 marks] Give a probabilistic expression for the number replaced by `(C)`. Note: You may use x to denote the number replaced by `(A)`, if you wish.

$2(1 - P(T \leq x))$ where $T \sim t_{24}$

Note: The degree of freedom is $n - 1 = 24$. If you had the incorrect degree of freedom you should only have lost 0.5 marks.

You are told that the value replaced by `(C)` is 0.02904.

(d) [2 marks] Based on the information provided in this question, would a 95% confidence interval contain 0? Carefully justify your answer. (Note: This question is asking if it would contain 0, it is <u>not</u> asking if it would contain 0.1.)

No. A 95% confidence interval would not contain 0.1 because $p < 0.05$. Furthermore, because the the sample mean is greater than 0.1, this means the lower bound of the 95% confidence interval must exceed 0.1 and, by extension, exceed 0.

Note: While we can say that a 95% confidence interval will not contain 0.1, it was necessary to carefully explain why it could not contain 0 either. This kind of question is designed to allow you to demonstrate how multiple parts of an analysis can be combined to reach a conclusion (in this case, knowledge of the p-value and sample mean, and how they relate to the confidence interval).

(e) [2 marks] Write down a probabilistic expression for the p-value resulting from a test of the null hypothesis $H_0 : \sigma^2 = 0.25$.

$2P(U \leq u)$ where $u = \frac{(n-1)s^2}{0.25}$, $n = 25$, $s = 0.289478$, and $U \sim \chi^2_{n-1}$.

Note: $2P(U \geq u)$ if $P(U \geq u) < 0.5$, otherwise $2P(U \leq u)$ was also acceptable, but in the above we know we can be more specific because $s^2 < 0.25$.

End of Examination